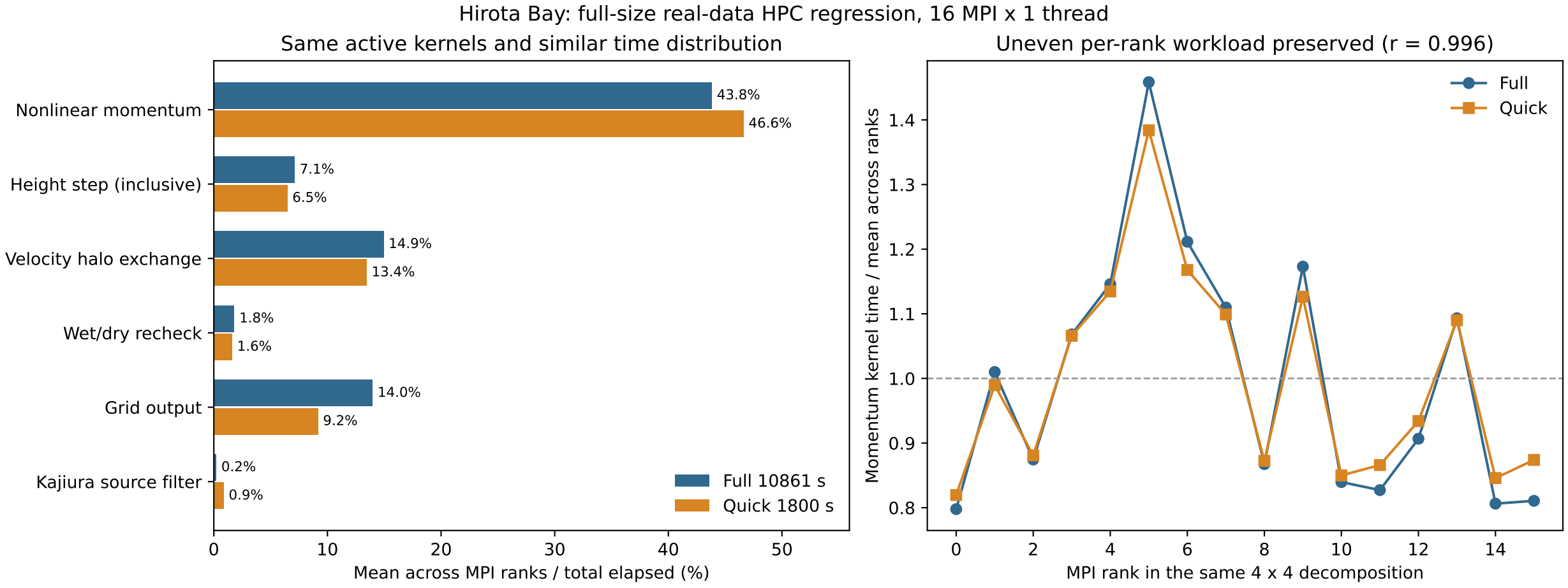
Hirota Bay: full-size real-data HPC regression, 16 MPI x 1 thread
Same active kernels and similar time distribution
Nonlinear momentum
Height step (inclusive)
Velocity halo exchange
Wet/dry recheck
Grid output
Kajiura source filter
43.8%
46.6%
7.1%
6.5%
14.9%
13.4%
1.8%
1.6%
14.0%
9.2%
0.2%
0.9%
Full 10861 s
Quick 1800 s
0
10
20
30
40
50
Mean across MPI ranks / total elapsed (%)
Uneven per-rank workload preserved (r = 0.996)
Full
Quick
Momentum kernel time / mean across ranks
0.8
0.9
1.0
1.1
1.2
1.3
1.4
0
2
4
6
8
10
12
14
MPI rank in the same 4 x 4 decomposition
Timer regions can overlap; percentages must not be added. Source work occupies a larger fraction of the short case.